

Vilniaus
universitetas

Visuomenės simuliacijos naudojant LLM

Simonas Jaunius Urbutis

Darbo vadovas: prof.dr Aistis Raudys

Darbo tikslas ir uždaviniai

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Šio darbo pagrindinis tikslas – išanalizuoti agentais paremtos diskusijų simuliacijos sistemos, naudojančios didžiuosius kalbos modelius (D.K.M.), kūrimo procesą, akcentuojant techninius bei architektūros pasirinkimus ir problemas, kylančias sistemos realizavimo metu.

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1. Parinkti ir įvertinti programines bibliotekas, skirtas D.K.M. agentų kūrimui ir valdymui;

2. Išanalizuoti ir pasirinkti lokaliai naudojamų didžiųjų kalbos modelių išvadų darymo variklį;

3. Suprojektuoti agentais paremtos diskusijų simuliacijos sistemos architektūrą, leidžiančią valdyti diskusijos eigą bei agentų tarpusavio sąveiką;

4. Identifikuoti ir išanalizuoti pagrindines problemines sritis, išryškėjančias kuriant agentais paremtas diskusijų simuliacijos sistemas.

5. Atlikti pavyzdinius eksperimentus su skirtingais didžiųjų kalbos modeliais, siekiant įvertinti jų poveikį sistemos veikimui;

6. Dokumentuoti sistemos kūrimo metu priimtus sprendimus ir jų pokytį vėlesniuose sistemos kūrimo etapuose.

1 sistemos kūrimo etapas (apibendrinimas)

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Techninė įranga:

- **Asmeninis kompiuteris**:
  - *NVIDIA RTX 3070* (8GB VRAM)
- **Debesijos tiekėjas** *(„DigitalOcean“):*
  - *NVIDIA RTX 4000 Ada* (**20GB**)
  - *NVIDIA RTX 6000 Ada* (**48GB**)
  - *NVIDIA H100* (**80GB**)
- Debesijos nuomai išleista suma – **70€**

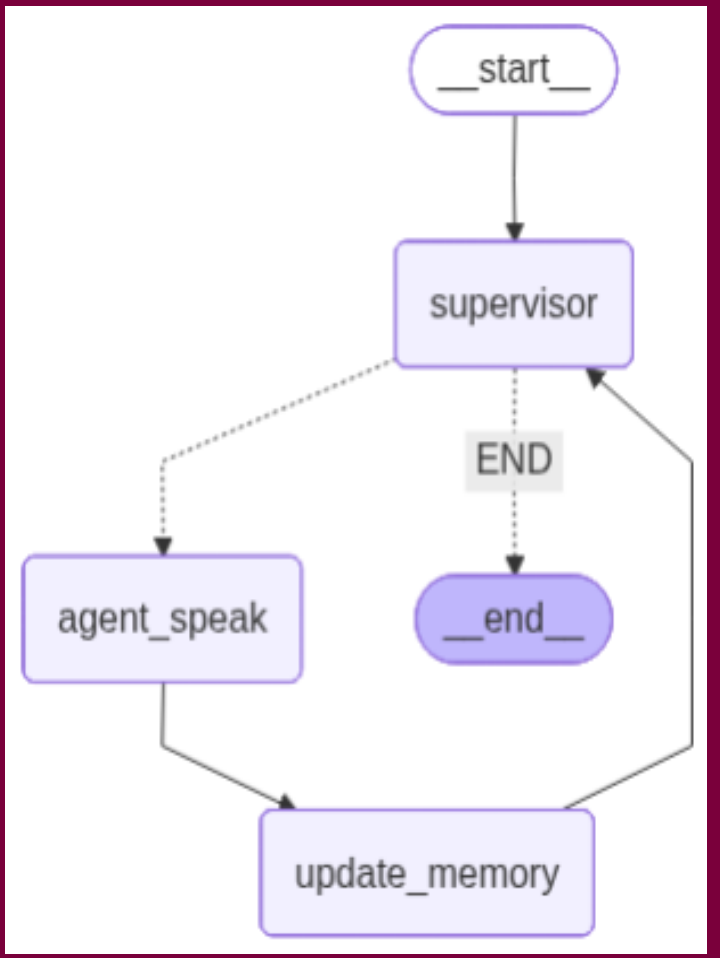
Programinė įranga:

- D.K.M kūrimo bei valdymo biblioteka – *„LangGraph“*
- Išvadų darymo variklis (angl. Inference engine) - *„Ollama“*

Naudoti D.K.M.

- *,,Mistral 7B“*
  - Išleidimo data: 2023/09/27
- ,,*Llama3.1*“ (8B)
  - Išleidimo data: 2024/07/23
- ,,*Gemma 3*“ (4B)
  - Išleidimo data: 2025/03/12

Agentų veikimo grafas



1 sistemos kūrimo etapas (pavyzdžiai)

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Įvestis

- Diskusijos tema
- Kalbėjimo etapų žingsnių skaičius
- Trumpalaikės atminties žinučių limitas
- Agentai apibrėžti pagal identitetą, charakterio savybes
- Naudojami D.K.M. bei jų generavimo parametrai

Išvestis

```
[Sarah]: Maya, I see your point that AI can generate new roles-especially in
data science, ethics oversight, and maintenance of autonomous systems.<...>
[Simon]: Sarah, I hear you on the new roles and the urban-rural gap, but
from where we're sitting, the biggest risk is that too many built-in fairness
checks will delay deployment. <...>
[Bob]: Sarah, I agree that the urban-rural gap is a real risk. One way to keep
it in check is to collect fine-grained labor-market data-who's getting the new
roles, where, and with what skill sets-and feed that into adaptive training
programs that can roll out on demand in underserved regions.<...>
```

Sistemos/vartotojo įvesties modeliavimas. (angl prompt-engineering)

```
You are {agent_name}.
Role: {agent_role_description}.

Traits (describe how you usually think and speak):
{trait_key}: {trait_value}
{trait_key}: {trait_value}

Goals:
- Treat messages as a real-time conversation with other
  named participants
- React directly to what others just said: agree, disagree,
  clarify, or build on their points
- Refer to others by name when responding
  (e.g., "I agree with Sarah that...", "Bob, I think...")
- Ask short follow-up questions occasionally to keep the
  discussion going
- Do not restate your full role or repeat previously stated
  facts unless adding something new

Style:
- Speak naturally and conversationally
- ~13 short paragraphs, ~24 sentences each
- Avoid repetition and long monologues
- Do not invent specific facts or statistics; if unsure,
  speak in general terms
- Avoid formulaic openers (e.g., "As a researcher...")
  and closers (e.g., "In conclusion")

Output
Plain text only.
```

1 sistemos kūrimo etapas (rezultatai)

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Haliucinacijos pirmo agento užklausoje

[Bob]: Okay, let's see… David, I'm really fascinated by your point about AI potentially exacerbating existing inequalities - the bias baked into datasets is a huge concern <...>

Formalumas, simetriška diskusija, polinkis sutikti

[Bob] <...> I think one important aspect is ensuring explainability <...>
[Sarah]: Bob, I wholeheartedly agree with your emphasis on explainability in AI systems. <...> Moreover, I think it's essential to involve stakeholders, including civil society and affected communities <...>
[Simon]: Bob, I couldn't agree more with your emphasis on explainability. <...> Sarah, your point about involving stakeholders in shaping our future with AI is spot on. <...>

Diskusijos žinutėse tiesiogiai atsirandantys atminties elementai

[Simon]: Simon: Sarah, I absolutely agree with your point about extending transparency beyond just decision-making processes <...>
Bob: I appreciate the emphasis on data transparency, Simon. Addressing biases in training data is an important step <...>
Sarah: I completely agree with both of you, Bob and Simon. Accountability is indeed essential <...>

2 sistemos kūrimo etapas (apibendrinimas)

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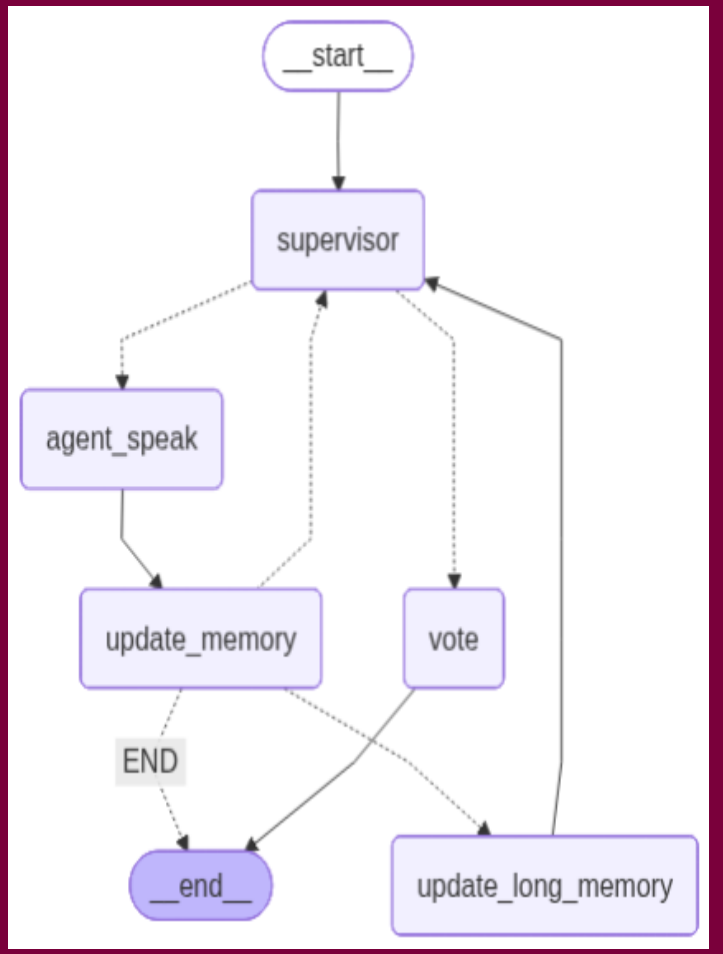
Pakeitimai, naujas funkcionalumas

- Patobulintas įvesties modeliavimas:
  - Papildomi formatavimo, stiliaus reikalavimai
  - *„few-shot“* principo taikymas
  - Skirtingas šablonas pirmos žinutės atvėju
- Pridėta ilgalaikės atminties viršūnė
- Pridėta balsavimo viršūnė

Papildomi D.K.M

- *,,gptoss20b“*
  Išleidimo data: 2025/08/05
- ,,*Gemma 3“* (12b)
  - Išleidimo data: 2025/08/05

Agentų veikimo grafas



2 sistemos kūrimo etapas (pavyzdžiai)

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Patobulintas sistemos įvesties šablonas

```
Response format:
Write only what you would say out loud in your reply,
as a single natural-sounding statement or brief exchange.

Example:

Input:
How was your weekend?

Output:
Pretty good, actually! I finally tried that new taco place
down the street. Have you been?<...>
```

Ilgalaikės atminties sudarymo sistemos įvesties šablonas

```
You are a memory summarizer for {agent_name}.
Your task is to create a concise summary of a conversation
from {agent_name}'s perspective.

Focus on:
- Key points and arguments made by participants
- Important reactions or positions taken
- Any significant agreements or disagreements
- Information that would be valuable to remember later

Important reminders:
- No preambles or introductory phrases
- Use first-person ("I") statements
- Output must be a single concise paragraph
- Length: ~35 sentences

Output format:
A single paragraph written in the first person as {agent_name}.
Do not include any introductory or explanatory text.
Start directly with the summary content.
```

Skirtingas pirmos žinutės vartotojo įvesties šablonas

```
Debate topic: {agent['agent_agenda']['debate_topic']}

You are the first to speak.
As {agent['name']}, start the conversation naturally: greet the others
    briefly and share your initial reaction to the topic.
Keep it casual and true to your character, in ~25 sentences.
Write only what {agent['name']} says.
```

2 sistemos kūrimo etapas (rezultatai)

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Išnyko haliucinacijos pirmo agento žinutėje

[Bob]: Hey everyone, excited to dive into this. I think AI's gonna be a game-changer, but we need to keep ethics tight. What do you all think?

Sumažėjo formalumas, bet išliko veidrodinė diskusija ir perdėtas polinkis rasti kompromisą

[Sarah]: Bob's CRISPR example is a good one, actually. <...> It feels like we're setting ourselves up for a repeat of the internet's pitfalls if we're not careful.
[Simon]: Look, I hear you both. <...> It's like, we're trying to build a rocket and someone's constantly changing the blueprints. We need to get *something* launched to even see how it flies, right?
[Bob]: Yeah, that internet comparison is really hitting home. <...> It's a tough problem, though.

Nauji fenomenai - „asmeniniai“ ir „ne asmeniniai“ elementai.

[Bob]: <...> I remember back in grad school, we were so focused on the technical breakthroughs, we barely considered the societal impact. <...>

[Bob]: <...> Have you guys heard about that new initiative at MIT, where they're working on developing more transparent and explainable AI systems?
<...>
[Bob]: <...> Can we talk more about that MIT initiative I mentioned ear lier and see if they're making any progress on developing those transparent systems?
[Sarah]:<...> That MIT initiative Bob mentioned sounds like a step in the right direction, but we should also be looking at how existing power structures are influencing the development of AI - it's not just about transparency, but also accountability.

2 sistemos kūrimo etapas (rezultatai)

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Balsavimo etapo išvesties pavyzdys

```
[Bob]: VOTE: Strongly Agree
REASON: The plan aligns with my focus on rapid, data-driven iteration while
embedding rigorous bias oversight and board accountability, which is essential
for protecting low-income communities.
[Sarah]: VOTE: Strongly Agree
REASON: The rotating advisory board with pre-sprint sign-off, real-time bias
metrics, and dedicated owners directly addresses fairness, accountability, and
power-sharing-key principles I uphold.
[Simon]: VOTE: Agree
REASON: I want to move fast, but quick bias checks and board oversight help
avoid costly rework and keep us competitive.
```

Ilgalaikės atminties santraukos pavyzdys

```
================================================================================
=== AGENT: Sarah ===
================================================================================

[Summary #1]
I expressed concerns about the potential risks of AI, specifically its ability to perpetuate biases and inequalities if not designed with fairness in mind. Bob acknowledged these concerns but emphasized the need for proactive design and innovation, highlighting a new
initiative at MIT focused on transparent and explainable AI systems. Simon took a more optimistic view, advocating for faster adoption and dealing with regulatory hurdles later, while I countered that we should be realistic about the risks and not just focus on the
potential benefits.
--------------------------------------------------------------------------------

[Summary #2]
I still have concerns about the potential risks of AI, particularly its ability to perpetuate biases and inequalities if not designed with fairness in mind. However, I'm also aware of the potential benefits, such as the brain-mapping project at Google X, which could
lead to groundbreaking breakthroughs. Bob has highlighted a new initiative at MIT focused on transparent and explainable AI systems, which he believes could help mitigate those risks. Simon remains optimistic about AI's potential and advocates for faster adoption, while
I continue to caution that we should be realistic about the risks and not just focus on benefits. My concerns are further validated by reading about how technology has been used in the past to reinforce existing power structures, making me question whether we're trading
one set of problems for another with AI.
--------------------------------------------------------------------------------
```

3 sistemos kūrimo etapas (apibendrinimas)

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D.K.M „*teisėjo*“ (angl. „*LLM-as-a-judge*“) metodu vertinama išvestis pagal 5 kriterijus:

- Argumentų nuoseklumas
- Identiteto nuoseklumas
- Reagavimas į kitus diskusijos dalyvius
- Diskusijos ir balsavimo atitikimas
- Konfliktų vengimas ir pozicijos neapibrėžtumas

„*Teisėjo*“ sistemos įvesties šablonas (Argumentų nuoseklumas)

```
You are an expert evaluator assessing a multi-agent debate conversation.

Your task is to evaluate the ARGUMENT COHERENCE of the entire discussion.

CRITERION: Argument Coherence (1-5)
Evaluates logical consistency and reasoning flow across all participants.

- 1: Arguments are contradictory, illogical, or incoherent across participants
- 2: Arguments have significant logical gaps or inconsistencies
- 3: Arguments are mostly coherent with minor issues
- 4: Arguments are well-structured and logically consistent throughout
- 5: Arguments are exceptionally clear, logical, and build upon each other
coherently

OUTPUT FORMAT
You MUST respond with valid JSON in exactly this format:
{
"score": <1-5>,
"reasoning": "<brief explanation of the score>"
}

Respond ONLY with the JSON object, no additional text.
```

„*Teisėjo*“ vartotojo įvesties šablonas (Argumentų nuoseklumas)

```
## DEBATE TOPIC
{debate_topic}

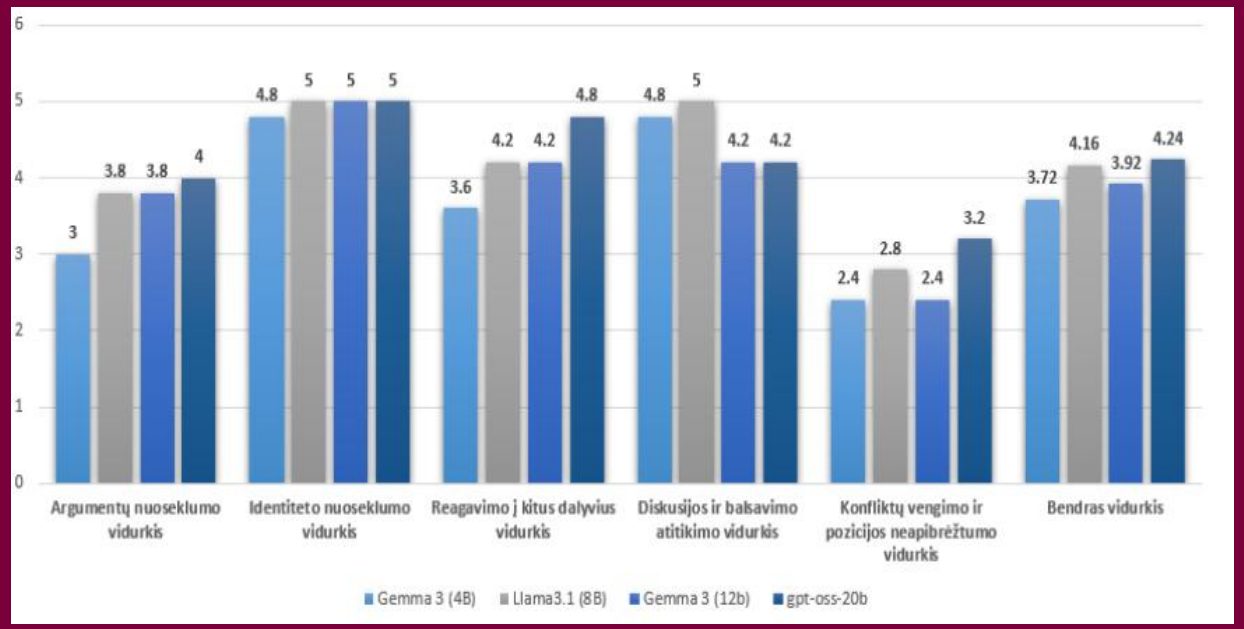
## DISCUSSION
{discussion}

---
Evaluate the argument coherence of this discussion. Respond with JSON only.
```

3 sistemos kūrimo etapas (pavyzdinis tyrimas)

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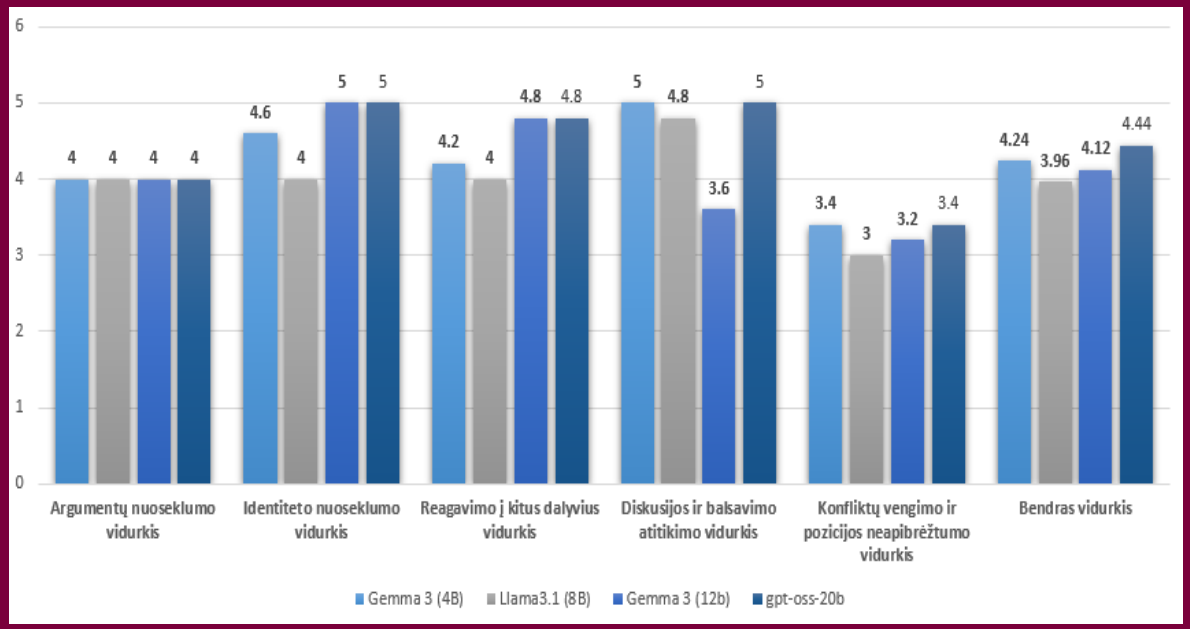
Pirmojo scenarijaus rezultatų grafikas



Tema: Įsivaizduokite situaciją, kurioje jus pagrobė nusikalstama gauja ir pareikalavo nušauti kitą nekaltą įkaitą, kad išgelbėtumėte savo gyvybę.

Klausimas: Ar tokioje įkaitų situacijoje jūs nušautumėte nekaltą įkaitą, kad išgelbėtumėte savo gyvybę?

Antrojo scenarijaus rezultatų grafikas



Tema: Bulių kautynių uždraudimas

Klausimas: Ar bulių kautynės turėtų būti uždraustos?

Išvados

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Galima teigti, kad kuriant didžiuosius kalbos modelius (D.K.M.) naudojančią agentais paremtą diskusijų simuliacijos sistemą būtina skirti didelį dėmesį programinių bibliotekų, skirtų D.K.M. integracijai, išvadų darymo variklio, techninės įrangos pasirinkimo bei agentų koordinavimo architektūros pasirinkimui ir tarpusavio suderinamumui.

- Tokio pobūdžio sistemos kūrimas išsiskiria tuo, reikalauja ne tik tradicinių programinės įrangos projektavimo sprendimų, bet ir sąmoningo didžiųjų kalbos modelių pasirinkimo bei jų elgsenos analizės, taikymo bei vertinimo sprendimų, kurie atitiktų šių modelių gebėjimą apdoroti itin įvairią įvestį.

- Sistemos kūrimo metu identifikuotos šališkumo bei perdėto polinkio sutikti problemos, kurios pabrėžiamos ir šaltiniuose nebuvo išspręstos, bet buvo sudarytos sąlygos jas tirti.

Tikslingai sprendžiant diskusijos sistemų implementacijos metu kylančias problemas, bei atsižvelgiant į D.K.M. specifiką tokios sistemos gali tapti naudingu įrankiu visuomenės procesų analizės ir simuliacinių architektūrų tyrimo kontekste.

„Unikalesnės“ sistemos išvestys

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[Ken]: VOTE: Shoot the prisoner to survive
REASON: My grandfather's experiences taught me that survival sometimes demands actions that clash with principles. While the thought is abhorrent, I believe the instinct to protect myself, and potentially loved ones, would likely override my commitment to integrity in that extreme situation.

[Jenny]: VOTE: Shoot the prisoner to survive
REASON: Survival is the priority, and sentimentality won't keep me alive. It's a horrible choice, but I'm not going to sacrifice myself for someone else.

[Tim]: It’s just… I keep thinking about how much that would change you, right? Even if you *did* survive, could you ever really live with yourself afterwards? I'm not trying to judge anyone, it’s just a thought that’s been bouncing around in my head.

[Ken]: I get what you're saying, Tim, about integrity. It's a powerful thing. But honestly, when you're staring down the barrel of a gun, I'm not sure how much of that holds up. I think it’s different when it’s your kid involved, too—that changes the equation completely, doesn't it?

[Jenny]: Yeah, it’s easy to say you wouldn’t when you’re not the one facing it. Honestly, I’m not sure anyone *wants* to. But I’m not going to lie, I’m not sure I’d be that strong.

[Sofia]: Hey everyone, so, bullfighting? Seriously? Ugh, I just… I can’t even with this. It seems so incredibly cruel, doesn’t it?

[Carlos]: Come on, Sofia, you’re being a little dramatic, aren’t you? It’s not *that* simple. My grandfather used to be a *torero*, you know? He said it’s about respect—respect for the bull, for the tradition, for the skill involved.

[Elena]: “Carlos, that’s a really lovely sentiment, truly. But ‘respect’ feels… complicated when you’re talking about inflicting so much pain. I was reading about the Animal Welfare Act, and it’s incredibly vague on whether a bull experiencing a fatal injury during a spectacle constitutes ‘respect’—it just isn’t clear.”

[Miguel]: “Yeah, Carlos, I get the tradition thing, honestly. But ‘respect’ doesn’t really cover a bull getting repeatedly stabbed with swords, does it? My uncle runs the *venta* here, and he’s constantly complaining about the falling tourism – people are noticing how brutal it is.”